Project Proposal: Bayesian Estimation

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1 Project Overview

Our group will be concerned with understanding, briefly educating, and producing a short report on Bayesian estimation. The framework assumes that parameters of interest are random with probability distributions that describe their uncertainty conditional on the observed dataset. It also treats the dataset as fixed, unlike a frequentist framework where the parameters are fixed and the data is assumed to be random and a result of repeated sampling from a larger population.

Our computational method of choice for Bayesian estimation is the Gibbs sampling method, which is a part of the Markov Chain Monte Carlo family. MCMC acts as the engine for Bayesian estimation, since it's not always possible to mathematically calculate the posterior distribution. Gibbs sampling is useful when the joint probability distribution $P(X, Y)$ is difficult to sample from directly, but sampling from the conditional distributions, $P(X|Y)$ and $P(Y|X)$ is feasible. We will start with a simple normal model that has known theoretical properties, and we want to see if our computational sampling method is approaching the expected distribution, in a sense we're looking to see if the history of the random walk traces out the probability distribution of interest. We then plan to explore more advanced models.

The spirit of the project is to quantify how accurately these computational methods help us converge the posterior.

1.1 Objectives

1.1.1 Simulation Study

We start of with a straightforward example using a normal model and we see how our simulation results compare to the theoretical results. After establishing the basic framework we extend the application to real data, then we explore how we can extend the methodology for hierarchical models.

1.1.2 Evaluating the Posterior

We will assess the means, variances, and credible intervals of posterior means, and discuss how simulation error and sample size influence our results. We will also investigate the convergence of our estimated posteriors using three methods: the Gelman-Rubin statistic $\sqrt{\hat{R}}$, the gibbs stopper method, and random initialization.

1.2 Background and Literature

2 Proposed Computational Strategy

Simple Example

Goal

We will start with a simple example to ensure we can correctly converge to some posterior. Alongside this, we will ensure we can calculate (and infer) posterior summary statistics, suitability of the model, monte-carlo standard error, and rate of convergence.

Methodology

This follows the example C.3 regarding the normals in Bayesain Data Analysis (BDA) Gelman et al. (2014).

1. Generate bivariate normal data
2. Derive and make functions for the marginal distributions (assuming normal prior and conditional)
3. Run the posterior simulations with Gibbs Sampling
4. Find the parameters through the samples
5. Repeat for various realistic n

Harder Example

Goal

Apply what was learned in the simple example to real data. We will be able to try and compare the our methodology.

Methodology

This follows the 1.6 example in BDA Gelman et al. (2014) but we turn it computational.

- Derive the full conditional, then the marginal conditionals using the distributions provided in text
- Gibbs sample for the posterior
- Explore the results
- Repeat for various realistic n

Extend to Hierarchical Models

Quantities of Interest

- Estimated mean, variance/standard deviation, credible intervals (informs us of the posterior model)
- Gelman-Rubin Statistic (Convergence Assessment)
- Rate of Convergence (computational efficiency)

Required Libraries

- MASS for the bivariate normal generation (or basic R) Ripley and Venables (2009)
- STAN for the actual MCMC/Gibs sampling – we may some examples by hand Guo et al. (2015)

3 Division of Responsibility

4 Timeline

We can chat about GANTT charts on tuesday or smth, just so we can layout realistic dates with buffer room.

5 Expected Outcomes

References

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2014). *Bayesian data analysis*. Texts in statistical science. CRC Press, Taylor & Francis Group, Boca Raton, third edition edition.

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